



ENEL301 Fundamentals of Engineering
Economics and Management

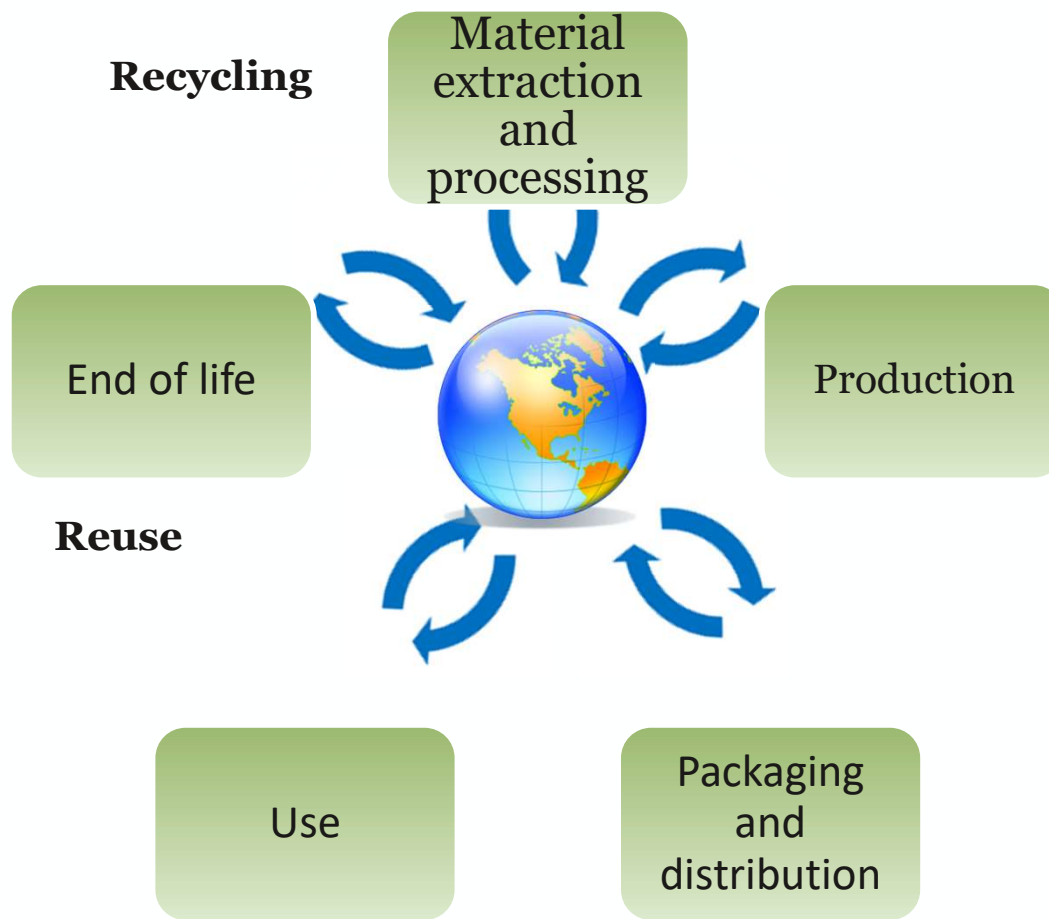
Life cycle assessment #1

Assoc. Prof. Enda Crossin
Director of MEM Programme

Introduction to LCA

- A tool to evaluate the environmental performance of a product or service
 - Society, **Environment**, Economy
- Framework: ISO 14040:2006 and ISO 14044:2006
- Can be used to assess social impacts (still being developed)
- NOT life cycle costing (although LCC can be done in parallel with LCA)

Introduction to LCA



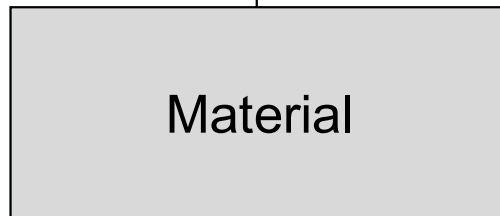
**Quantify the environmental impacts over the full life cycle....
Product or service**

Impact category	Typical units
Global warming (Carbon)	kg CO ₂ -eq
Embodied energy	MJ
Acidification	kg SO ₂ -eq
Eutrophication	kg PO ₄ ⁻³ -eq
Ecological footprint	Ha.a
Water use	kL
Carcinogens	DALY (Disability adjusted life years)



Introduction to LCA

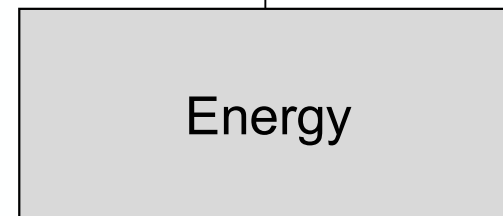
Emissions to air,
water and ground (e.g. NO_x)



Material

Raw materials and resources
(e.g. iron ore, land, water)

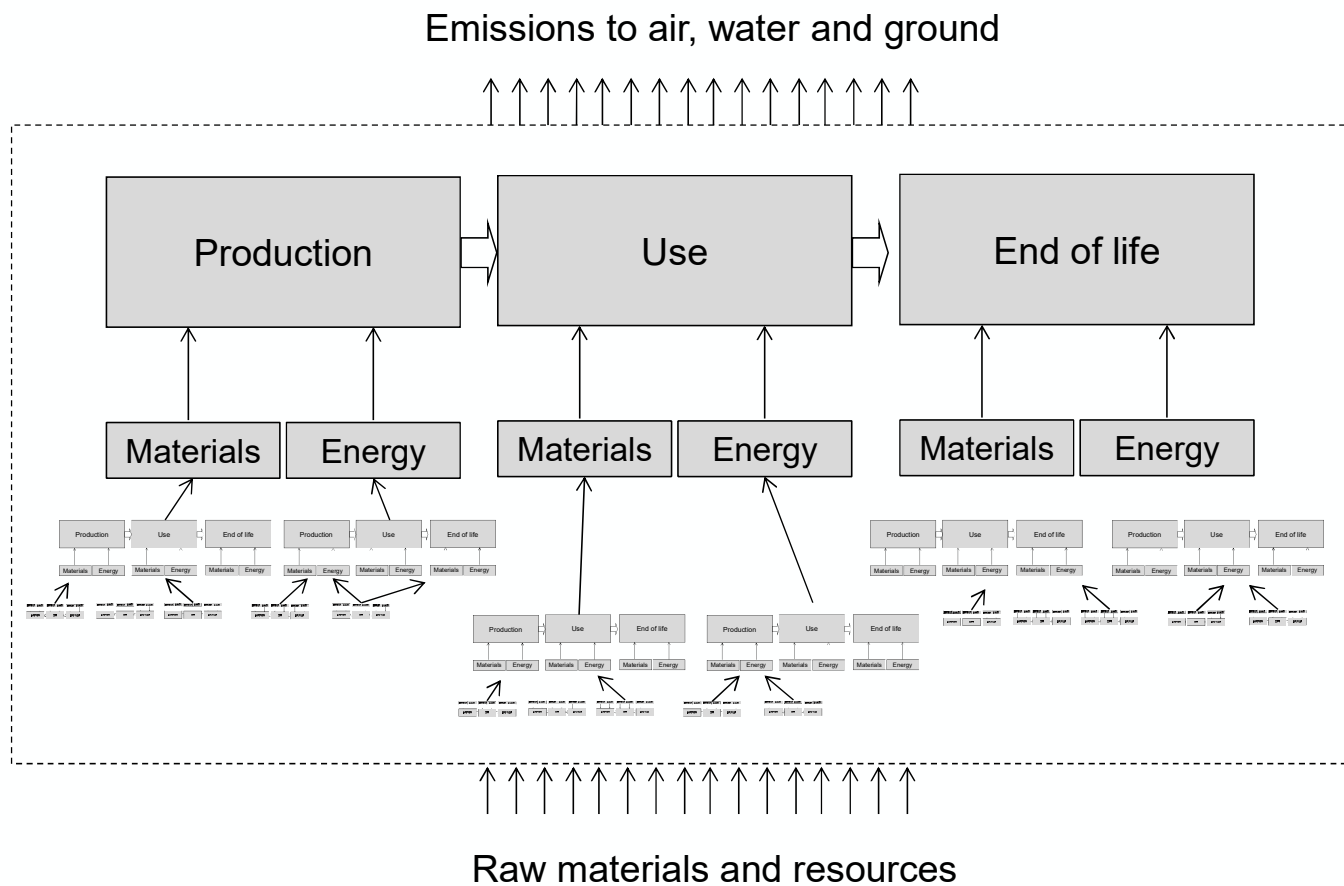
Emissions to air,
water and ground (e.g. CO₂)



Energy

Raw materials and resources
(e.g. natural gas)

Introduction to LCA



Introduction to LCA

“Classic” LCA questions

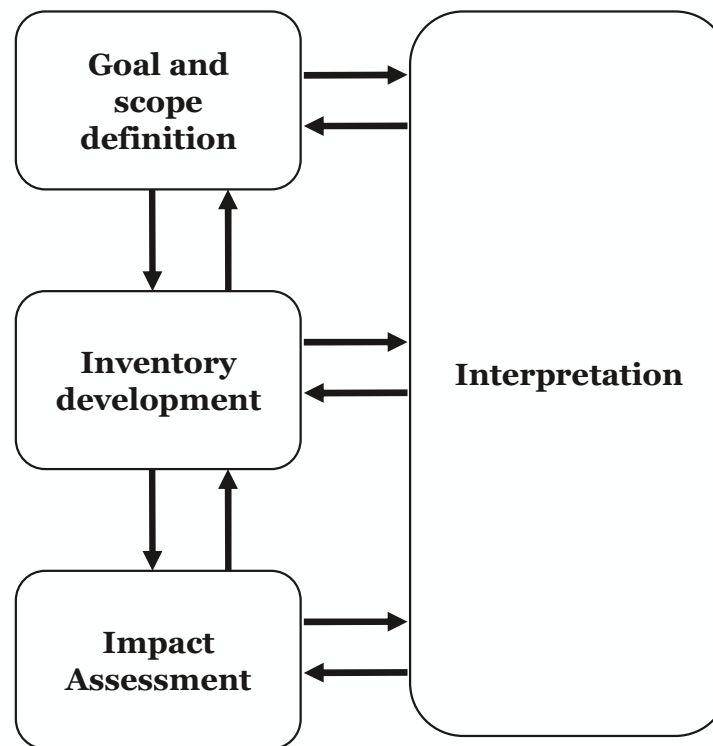
- Drinking containers: Glass vs. PET vs. Aluminium
- Disposable vs cotton nappies
- Electric car vs petrol car
- Does the embodied energy in a solar panel ever pay for itself?

Introduction to LCA

Why?

- Design tool
- Product development and improvement
- Strategic planning
- Public policy development
- Marketing
- Other

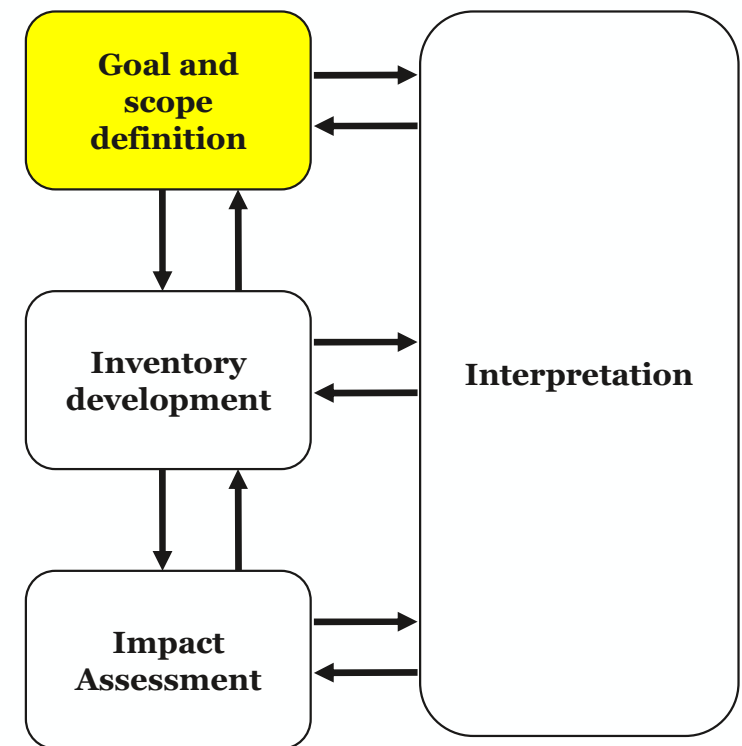
How to do an LCA



Source: ISO 14040:2006

Goal and scope definition

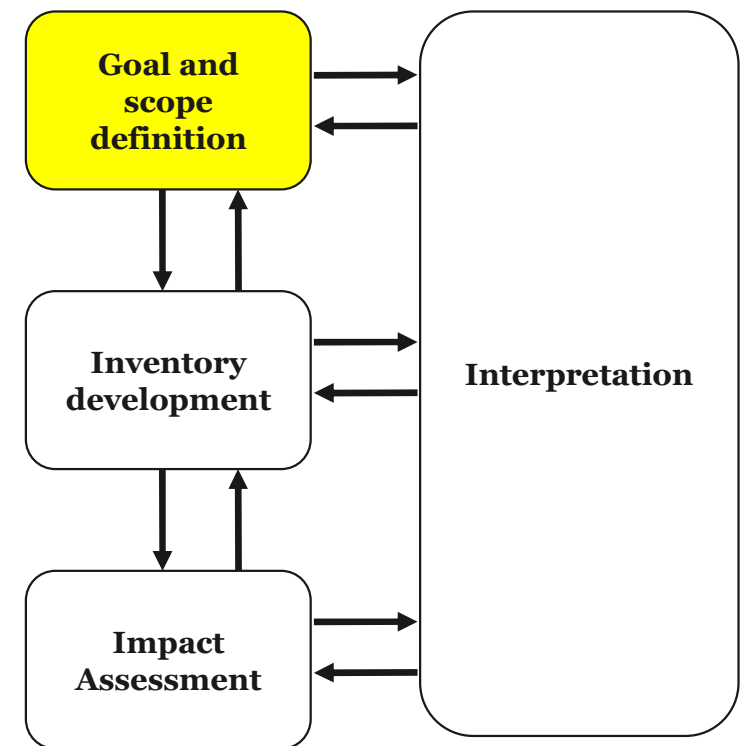
- Frame the study
- Functional unit
- System boundary
- Identify environmental indicators
- Identify data requirements



Source: ISO 14040:2006

Frame the study

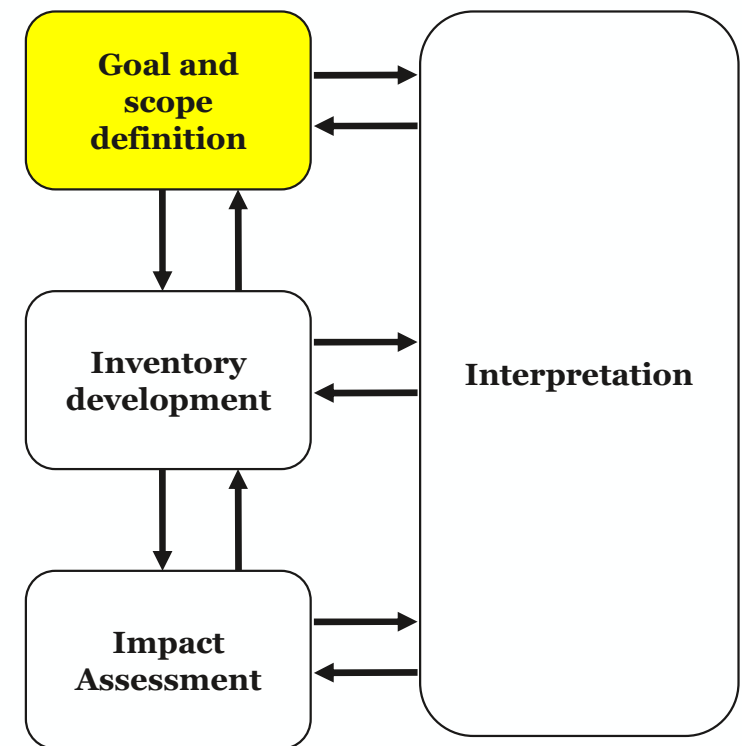
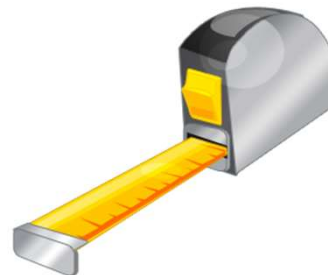
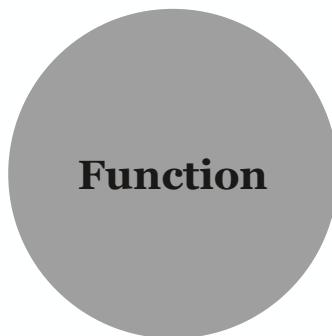
- Reasons for study & objectives
- Key methodological decisions
- What is included/excluded)
- Audience; internal, external, marketing purpose (if any)



Source: ISO 14040:2006

Functional unit

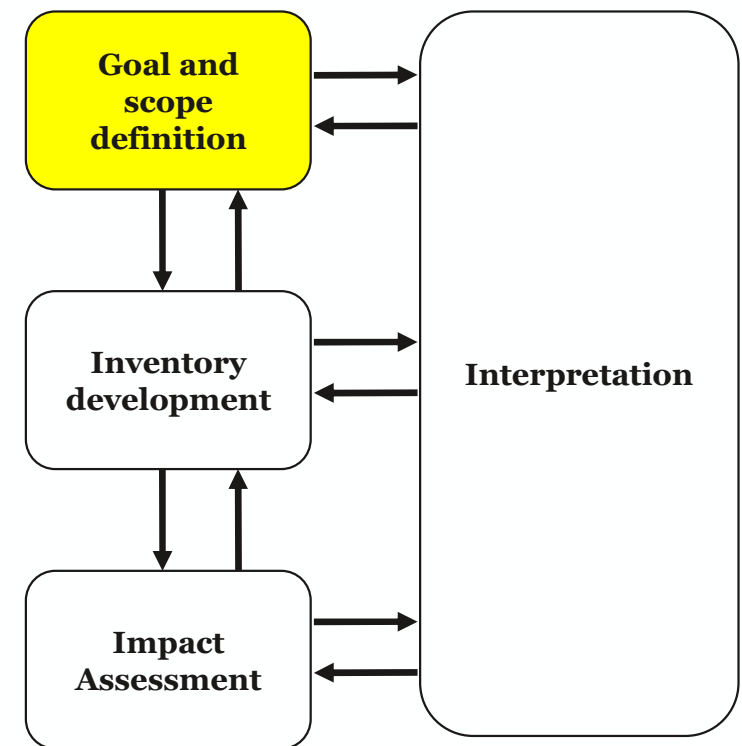
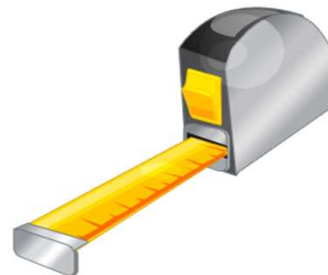
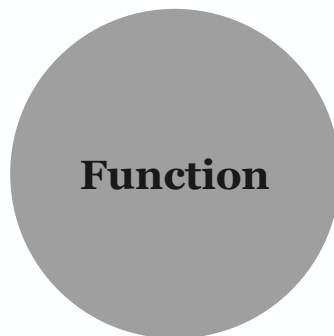
- Identify the primary function
- Quantify that function
- Allows for fairer comparison between systems



Source: ISO 14040:2006

Functional unit

- Soft-drink packaging: Glass vs. PET vs. Al
Functional unit: Delivery of 1 L of soft drink to a consumer
- Electric car vs petrol car
Functional unit: Move one person one kilometer (person.km)
- Does the embodied energy in a solar panel ever pay for itself?
Functional unit: 1 kWh of electricity to a household



Source: ISO 14040:2006

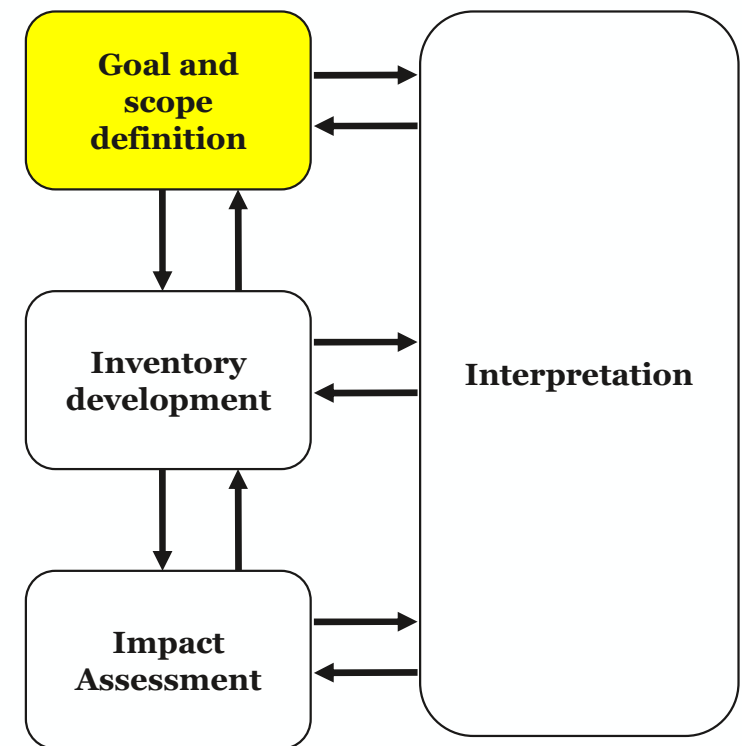
System boundary

“...which ... processes are part of a product system” (ISO 14040:2006)

- Attempt to cover 95% + of environmental impacts associated with the product life cycle
- How do you know what is remaining?

Typical exclusions:

- Capital equipment and maintenance
- Human labour, except for labour-intensive
- Accidents



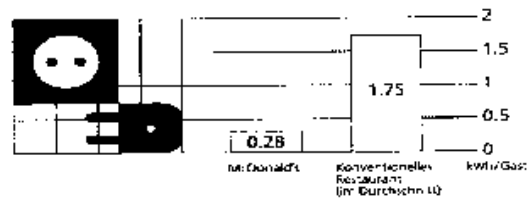
Source: ISO 14040:2006

System Boundary the McDonald's example

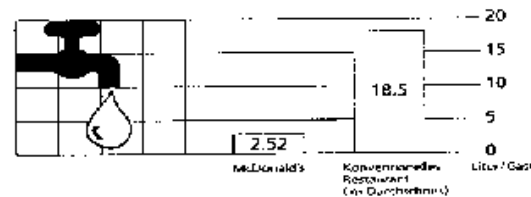
McDonald's ist umweltbewusst!

Der Beweis:

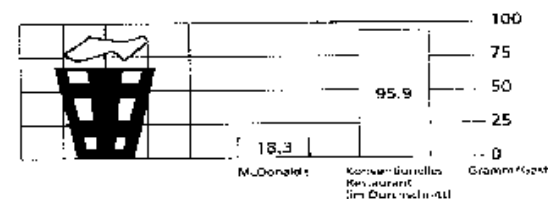
6 × weniger Energie!



7 × weniger Wasser!



5 × weniger Abfall!



McDonald's “proves”
its environmental
credentials:

6 x less energy

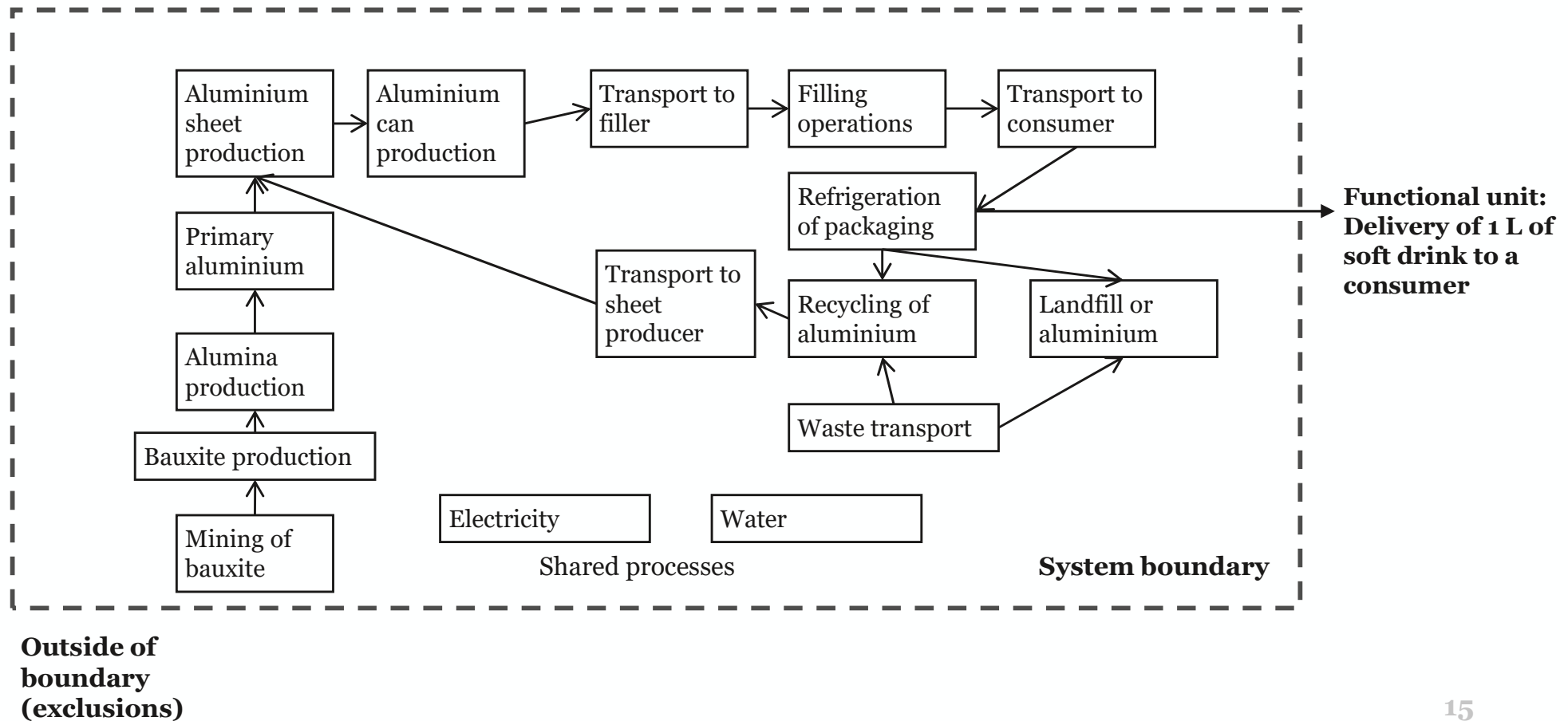
7 x less water

5 x less waste

compared to a
conventional restaurant

System Boundary – Classic LCA questions

Soft-drink packaging: Glass vs. PET vs. Al

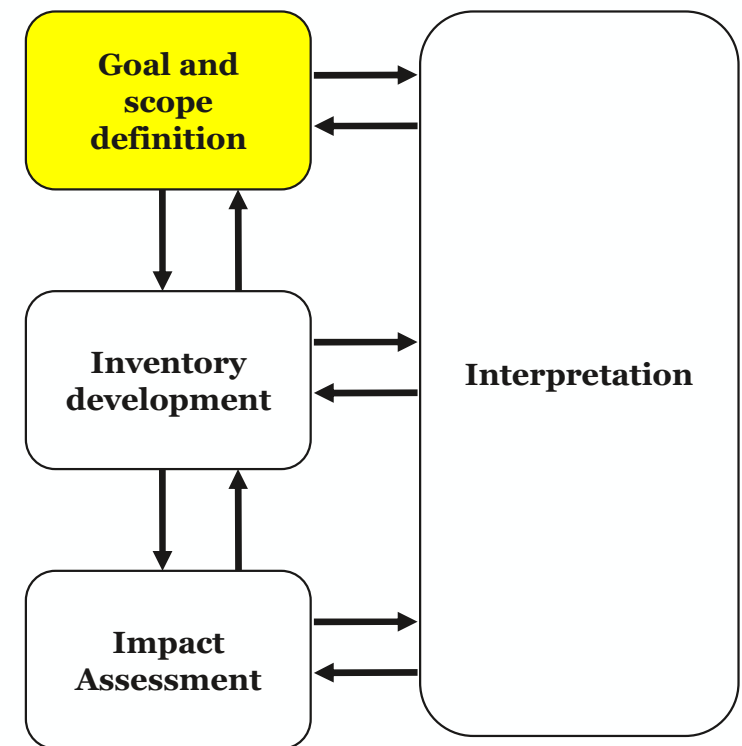


Identify environmental indicators

- What environmental indicators are you interested in studying?
- Needs to be relevant to the study
- Don't ignore known issues
- 30+ indicators you could consider

Where to start:

- What do stakeholders think are important?
- What indicators have been used elsewhere?
- Best practice guides are available



Source: ISO 14040:2006

Identifying indicators

Climate change potential

- Air emissions of CO₂, CH₄, N₂O, SF₆, CFCs etc
- Often tracks energy from fossil fuels, so can be a good proxy for many other fossil fuel related impacts
- *Climate change potential, kg CO₂-eq*
- Carbon accounting methods, inc. time horizons can affect result



Image source:
www.celsias.com

Identifying indicators

Photochemical oxidation (smog)

- Air emissions of SO_x, NO_x, CO etc. etc.
- Photochemical ozone formation is caused by degradation of volatile organic compounds (VOC) in the presence of light and nitrogen oxide (NO_x).
- *Photochemical ozone creation potentials POCP*, kg C₂H₄-eq (ethylene equivalents)



Image source:
www.travelersdigest.com

Identifying indicators

Acidification

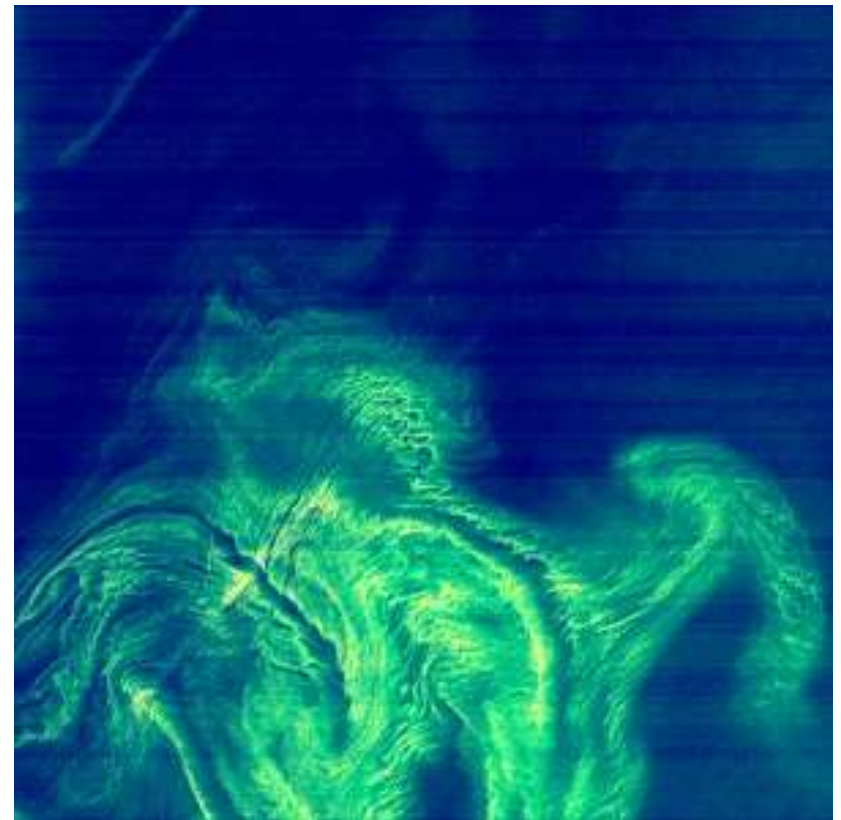
- Air, soil and water emissions of SO_x, NO_x, HF, etc. etc.
- Release of acids or acid-forming compounds in the terrestrial (soil) or aquatic ecosystems.
- Acid rain is a problem mainly in Scandinavian and European countries
- *Acidification potential*, kg SO₂-eq



Identifying indicators

Eutrophication/Nutrification

- Air, soil and water emissions of NO_x, PO_x etc. etc.
- Eutrophication of aquatic and nutrification of terrestrial ecosystems is caused by surplus nitrogen, phosphorus, and degradable organic substances. Contributes to algal blooms
- Eutrophication Potential, kg PO₄³⁻-eq (phosphate equivalents)



Identifying indicators

Land use

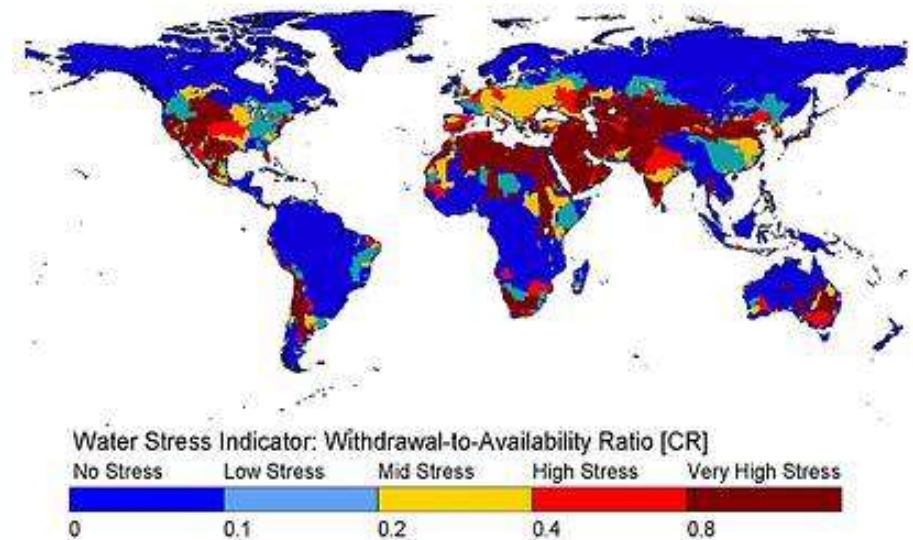
- Sum of land occupied over the life cycle
- Ha.a (occupation over time)
- A (poor) proxy for land use impacts
- Often coupled with soil quality, water use



Identifying indicators

Water use

- Sum of water inputs over the life cycle
- kL H₂O
- New methods being developed to account for water scarcity



Identifying indicators

Minerals and fossil fuel depletion

- Consumption of minerals and fossil fuels
- Measure of the pressure / strain on resource use
- Different extraction measures:
 - Additional energy
 - Additional investment



Identifying indicators

Human toxicity

- Air, soil and water emissions of dioxins, cadmium, arsenic etc. etc.
- **Disability Adjusted Life Years (DALY)**
 - A World Health Organisation measure also used in health funding to classify death and disability.



Image source:
www.francis.edu

Identify environmental indicators

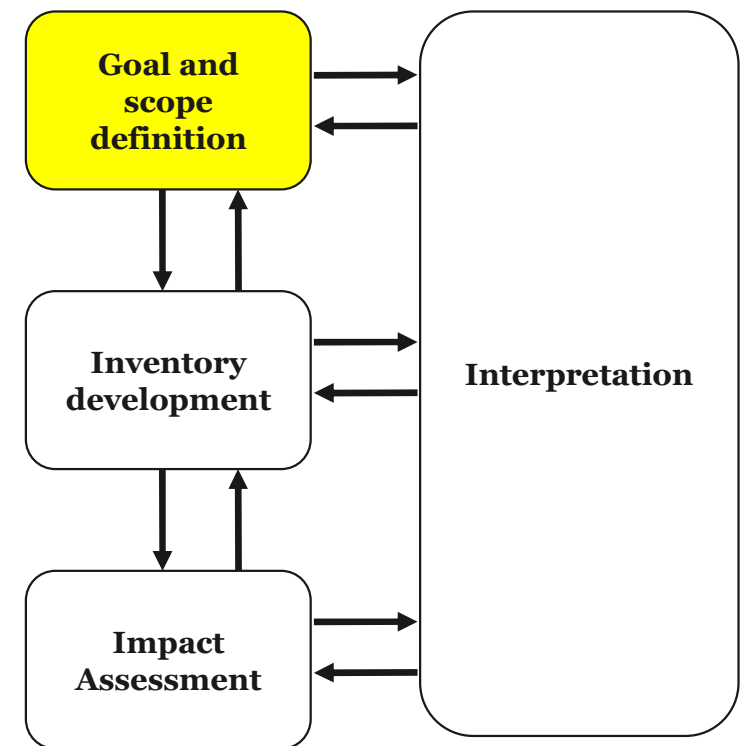
Impact factors

Impact category climate change - 100a

Flow	Category	Flow property	Factor	Unit	Uncertainty
Carbon dioxide	Emission to air/low population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, fossil	Emission to air/high population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, fossil	Emission to air/low population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, fossil	Emission to air/low population density, long-term	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, fossil	Emission to air/low stratosphere + upper troposphere	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, fossil	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, from soil or biomass stock	Emission to air/low population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, from soil or biomass stock	Emission to air/low population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, from soil or biomass stock	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, from soil or biomass stock	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, land transformation	Emission to air/high population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, land transformation	Emission to air/low population density, long-term	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, land transformation	Emission to air/lower stratosphere + upper troposphere	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, non-fossil	Emission to air/high population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, non-fossil	Emission to air/low population density	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, non-fossil	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon dioxide, non-fossil, from calcination	Emission to air/unspecified	Mass	1.0	kg CO2-Eq/kg	none
Carbon monoxide, fossil	Emission to air/high population density	Mass	1.57	kg CO2-Eq/kg	none
Carbon monoxide, fossil	Emission to air/low population density	Mass	1.57	kg CO2-Eq/kg	none
Carbon monoxide, fossil	Emission to air/low population density, long-term	Mass	1.57	kg CO2-Eq/kg	none
Carbon monoxide, fossil	Emission to air/lower stratosphere + upper troposphere	Mass	1.57	kg CO2-Eq/kg	none
Carbon monoxide, fossil	Emission to air/unspecified	Mass	1.57	kg CO2-Eq/kg	none
Dinitrogen monoxide	Emission to air/high population density	Mass	298.0	kg CO2-Eq/kg	none
Dinitrogen monoxide	Emission to air/low population density	Mass	298.0	kg CO2-Eq/kg	none
Dinitrogen monoxide	Emission to air/low population density, long-term	Mass	298.0	kg CO2-Eq/kg	none
Dinitrogen monoxide	Emission to air/lower stratosphere + upper troposphere	Mass	298.0	kg CO2-Eq/kg	none
Dinitrogen monoxide	Emission to air/unspecified	Mass	298.0	kg CO2-Eq/kg	none
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Emission to air/high population density	Mass	1430.0	kg CO2-Eq/kg	none
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Emission to air/low population density	Mass	1430.0	kg CO2-Eq/kg	none
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Emission to air/low population density, long-term	Mass	1430.0	kg CO2-Eq/kg	none
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Emission to air/lower stratosphere + upper troposphere	Mass	1430.0	kg CO2-Eq/kg	none
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Emission to air/unspecified	Mass	1430.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trichloro-, HCFC-140	Emission to air/high population density	Mass	146.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trichloro-, HCFC-140	Emission to air/low population density	Mass	146.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trichloro-, HCFC-140	Emission to air/low population density, long-term	Mass	146.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trichloro-, HCFC-140	Emission to air/lower stratosphere + upper troposphere	Mass	146.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trichloro-, HCFC-140	Emission to air/unspecified	Mass	146.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trifluoro-, HFC-143a	Emission to air/high population density	Mass	4470.0	kg CO2-Eq/kg	none
Ethane, 1,1,1-trifluoro-, HFC-143a	Emission to air/low population density	Mass	4470.0	kg CO2-Eq/kg	none

Goal and scope definition

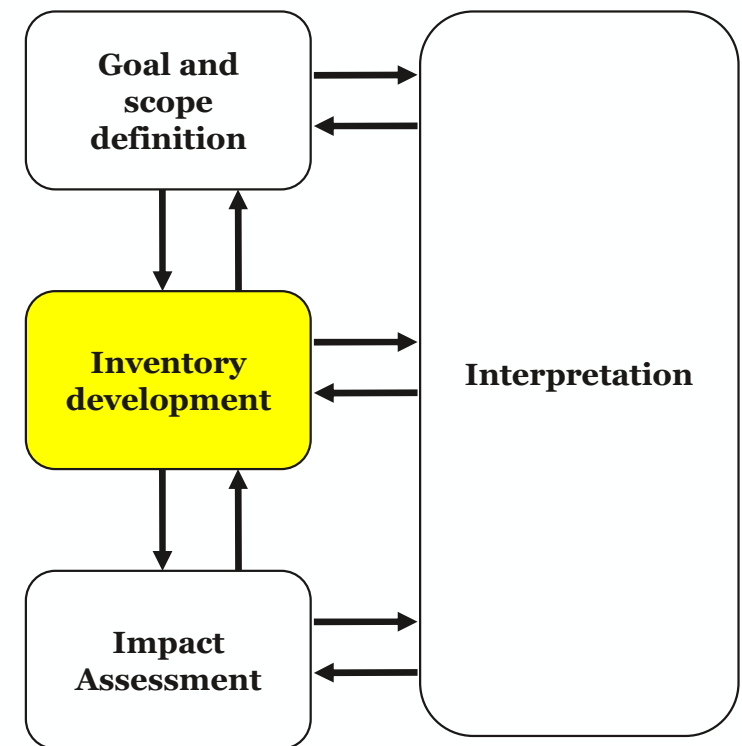
- Frame the study
- Functional unit
- System boundary
- Identify environmental indicators
- **Identify data requirements**



Source: ISO 14040:2006

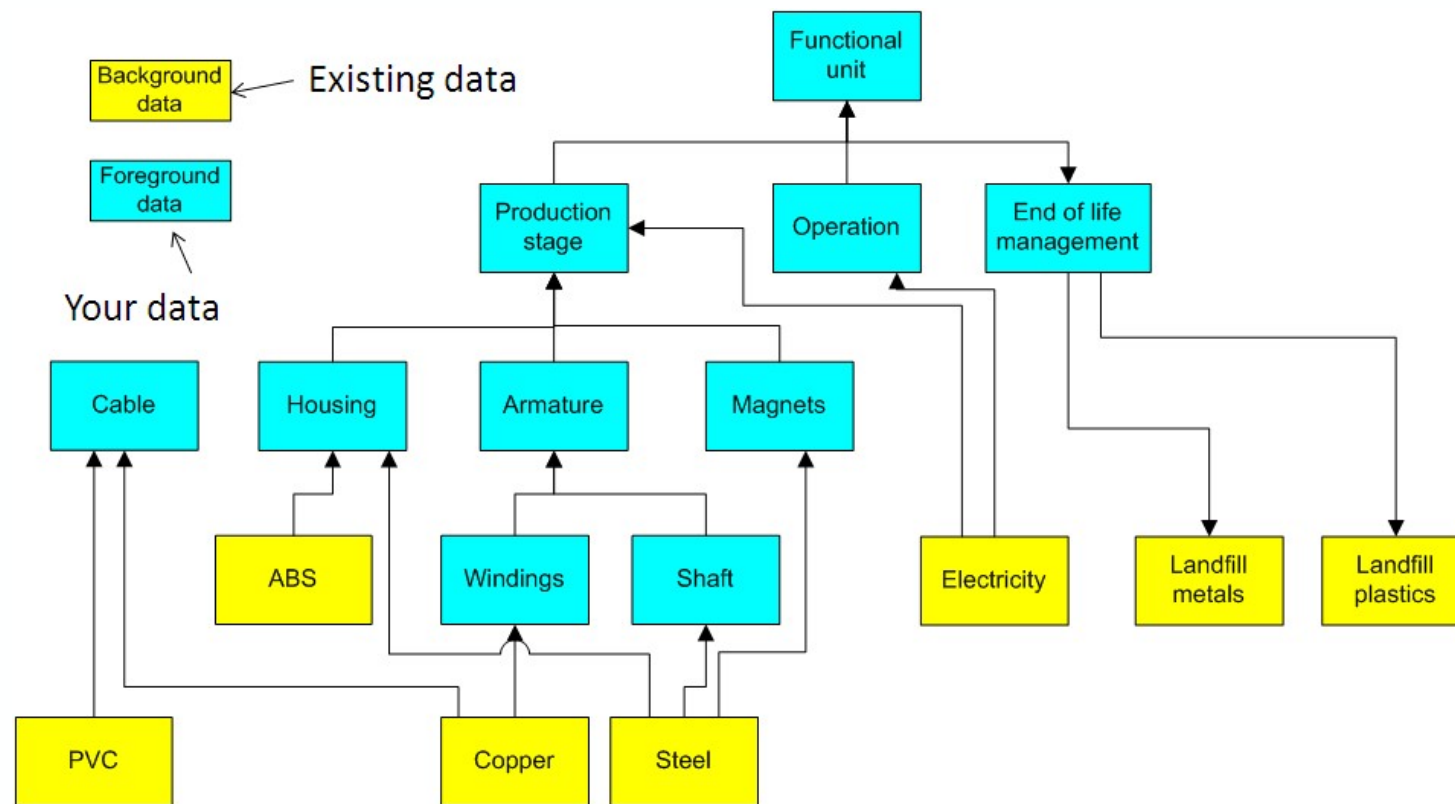
Inventory development

- LCA projects breath on data
- Foreground data
 - Data that the original manufacturer supplies, e.g. mass of packaging, energy inputs for a process, emissions
- Background data
 - Data that the LCA practitioner supplies
 - e.g. electricity generation, transport, production of common materials
 - Typically from LCA databases



Source: ISO 14040:2006

Inventory development – electric motor



Inventory development – example survey

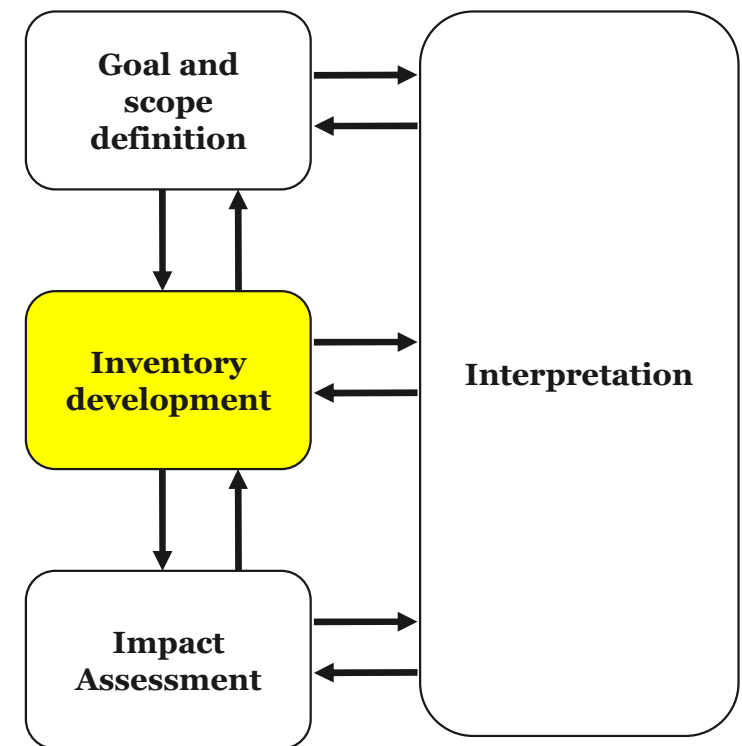
1 Raw Materials Inbound	Concrete	Brick	Rock	Asphalt	Total	
					-	tonnes or m ³
2 Electricity Usage	Production-Concrete	Other production	Administration	Other	Total	
					-	kWh
3 Plant Output	Recycled 20 mm concrete	Recycled 14 mm concrete	Other recycled concrete	Steel		tonnes
4 By products from Production [tonnes]	Plastics	Wood	Other waste		Total	
					-	
	Production	Dust Suppression				
5 Water Consumption					-	Mltrs
Water sources						
6 Dust produced around plant (approximated)					-	mg / m ³
7 Percentage of transport vehicles picking up inbound materials after 'dropping off' outbound materials i.e. backloading						
	On site plant Equipment-concrete	Transport of waste to site	Total transport of recycled concrete to concrete plants	Transport of recycled concrete to Boral North Melbourne	Total	
8 Diesel Consumption					-	kL
9 Transport types (e.g. articulated, rigid, dog-trailer)						
10 Transport distances					-	km
Notes						
Diesel Consumption-Site Total, includes various operations ie. transport, subdivision works, bin delivery etc						

Inventory development

- Document the inventory
- → makes your report **transparent**

Rule of thumb (like any method section):

- Enough information that someone should be able to pickup your report and repeat the result



Source: ISO 14040:2006

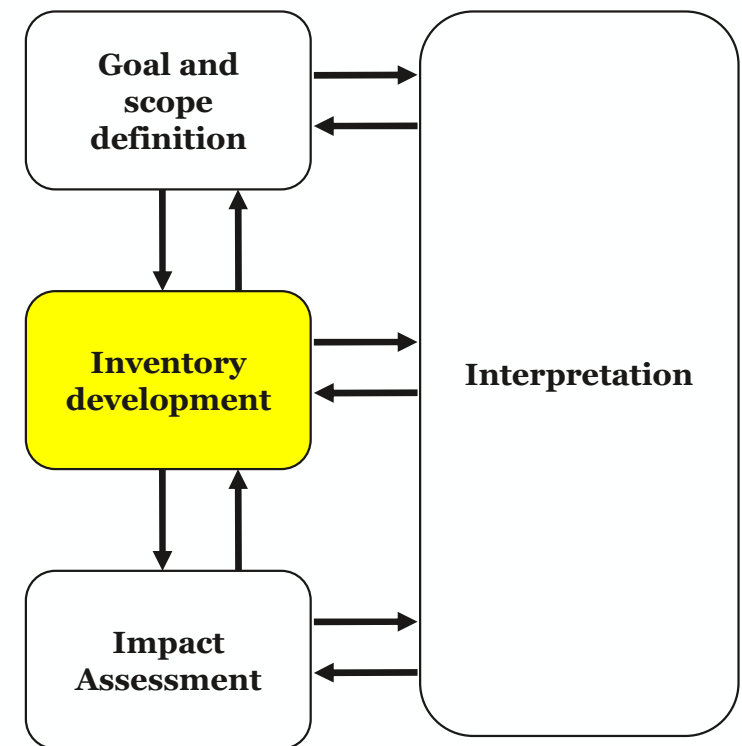
Inventory development

LCA practitioner

- Couples foreground and background data
- Runs a preliminary assessment
- Looks for data gaps/errors

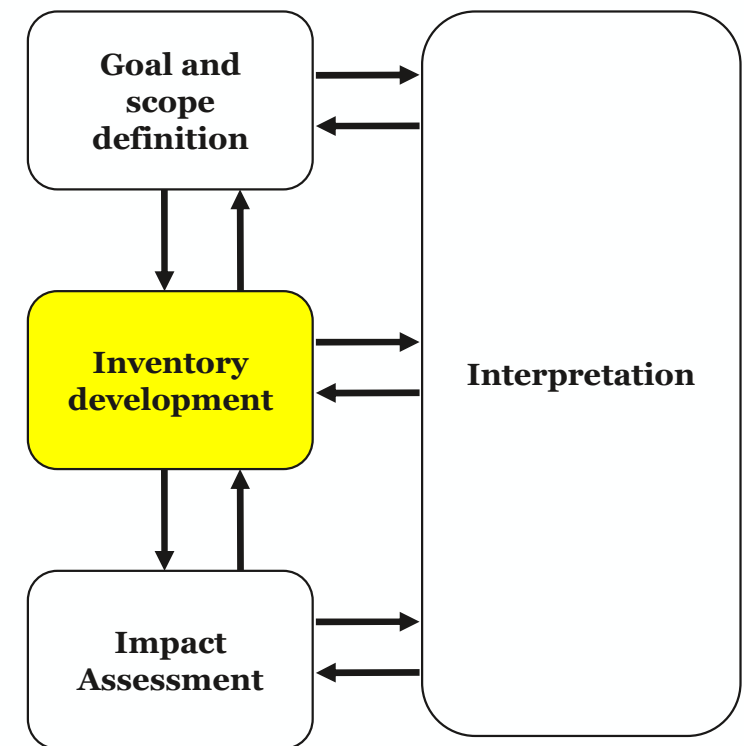
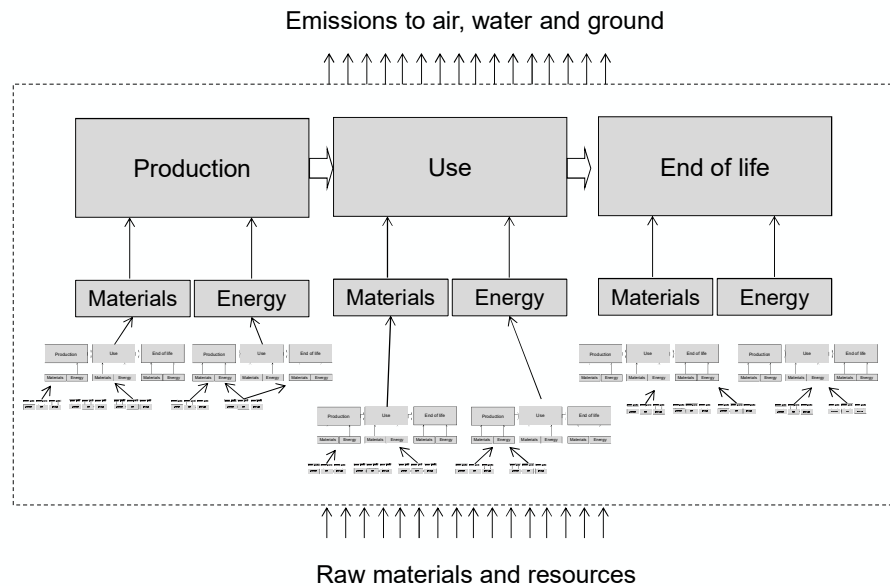
LCA commissioner

- Often likes to view inventory
- Provide more data to fill gaps/correct errors



Source: ISO 14040:2006

Inventory development



Source: ISO 14040:2006

Inventory development

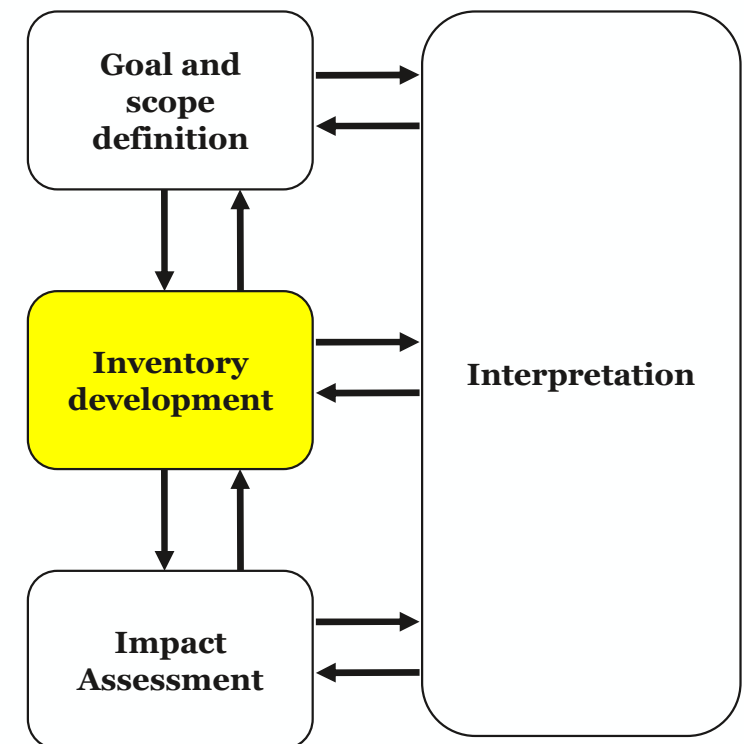
Inputs/Outputs: Mid-size trolley backgate, at IPS

Inputs

Flow	Category	Amount	Unit
aluminium, wrought alloy	242:Manufacture of basic precious and ...	0.10400	kg
electricity, low voltage, Victoria/AU U	Electricity country mix/Low Voltage	2.27000	kWh
HDPE at IPS	01_Project flows	1.17200	kg
injection moulding	222:Manufacture of plastics products/2...	1.17200	kg

Outputs

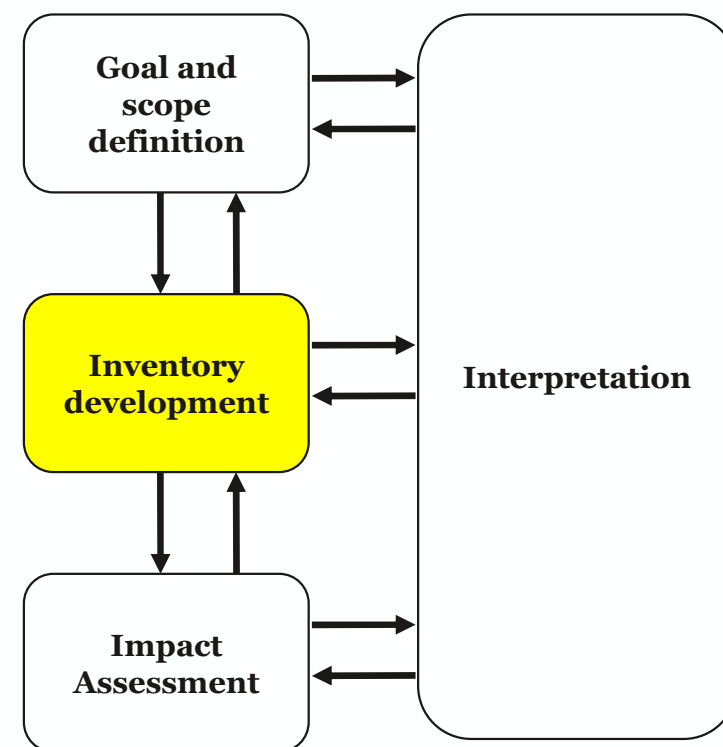
Flow	Category	Amount	Unit
Mid-size trolley backgate	01_Project flows	1.00000	Item(s)



Source: ISO 14040:2006

Inventory development

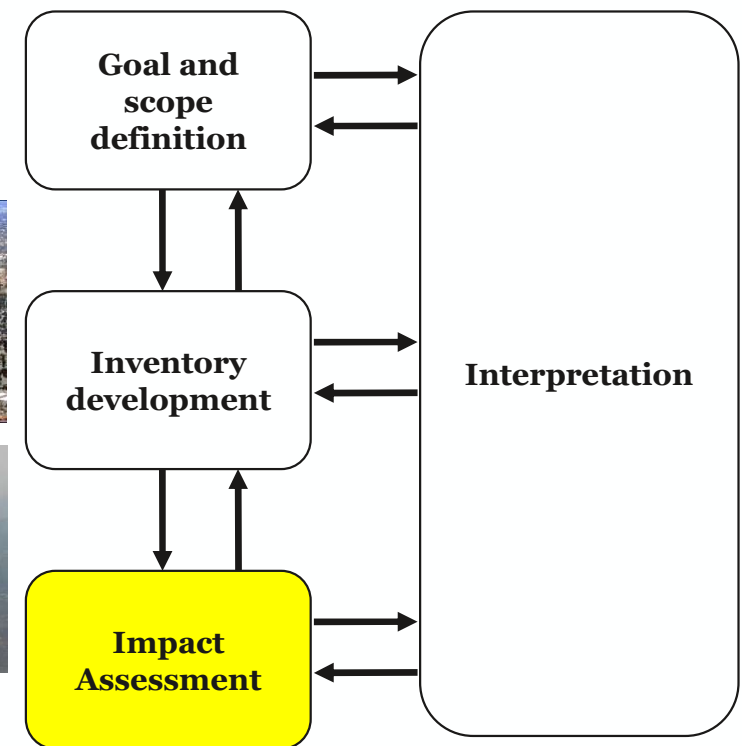
Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94	0.43	µg
Air	Raw	406.22	180.04	ng
Aluminum hydroxide	Raw	168.16	180.16	mg
Carbon	Raw	1.28	0.57	mg
Coal washery waste	Raw	0.00	56.95	g
Energy, from solar	Raw	5.76	2.55	kJ
Energy, kinetic, flow, in wind	Raw	0.00	193.63	J
Energy, recovered	Raw	0.00	0.00	J
Energy, unspecified	Raw	369.93	106.59	MJ
Ferromanganese	Raw	505.11	73.32	µg
Granite, in ground	Raw	1.14	0.00	tonne
Graphite, from technosphere, 50MJ/kg	Raw	221.81	32.20	ng
Occupation, industrial area	Raw	0.61	0.29	m2s
Phosphorus pentoxide	Raw	77.64	82.99	mg
Potassium chloride	Raw	12.23	5.42	ng
Secondary glass	Raw	115.43	19.29	µg
Sulfur dioxide, secondary	Raw	321.99	0.25	g
Waste, concrete	Raw	-1.44	-1.66	tonne
Waste, fly ash	Raw	75.00	170.00	kg
bagasse	Raw	29.39	13.57	g
bagasse, 8.7MJ/kg	Raw	572.88	83.16	mg
Energy, from biomass	Raw	4.53	2.01	MJ
etc. etc.				
Outflows				
Carbon dioxide	Air	269.79	230.95	kg
Carbon dioxide, biogenic	Air	256.60	114.33	g
Carbon dioxide, fossil	Air	50.43	22.45	kg
Carbon disulfide	Air	0.80	3.52	mg
Carbon monoxide	Air	379.78	451.32	g
Carbon monoxide, biogenic	Air	0.00	100.41	mg
Carbon monoxide, fossil	Air	0.00	1.42	g
Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00	0.01	pg
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Air	21.37	45.17	mg
Ethane, 1,2-dibromo-	Air	28.27	33.20	µg
Ethane, 1,2-dichloro-	Air	640.86	284.03	pg
1-Methylfluorene	Water	6.71	7.87	µg
2-Hexanone	Water	384.10	451.09	µg
2-Methylnaphthalene	Water	0.93	1.09	mg



Source: ISO 14040:2006

Impact assessment

Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94	0.43	µg
Air	Raw	406.22	180.04	ng
Aluminum hydroxide	Raw	168.16	180.16	mg
Carbon	Raw	1.28	0.57	mg
Coal washery waste	Raw	0.00	56.95	g
Energy, from solar	Raw	5.76	2.55	kJ
Energy, kinetic, flow, in wind	Raw	0.00	193.63	J
Energy, recovered	Raw	0.00	0.00	J
Energy, unspecified	Raw	369.93	106.59	MJ
Ferromanganese	Raw	505.11	73.32	µg
Granite, in ground	Raw	1.14	0.00	tonne
Graphite, from technosphere, 50MJ/kg	Raw	221.81	32.20	ng
Occupation, industrial area	Raw	0.61	0.29	m2s
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Secondary glass	Raw	115.43	19.29	µg
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Waste, fly ash	Raw	75.00	170.00	kg
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bagasse, 8.7MJ/kg	Raw	572.88	83.16	mg
Energy, from biomass	Raw	4.53	2.01	MJ
etc. etc.				
Outflows				
Carbon dioxide	Air	269.79	230.95	kg
Carbon dioxide, biogenic	Air	256.60	114.33	g
Carbon dioxide, fossil	Air	50.43	22.45	kg
Carbon disulfide	Air	0.80	3.52	mg
Carbon monoxide	Air	379.78	451.32	g
Carbon monoxide, biogenic	Air	0.00	100.41	mg
Carbon monoxide, fossil	Air	0.00	1.42	g
Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00	0.01	pg
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Air	21.37	45.17	mg
Ethane, 1,2-dibromo-	Air	28.27	33.20	µg
Ethane, 1,2-dichloro-	Air	640.86	284.03	pg
1-Methylfluorene	Water	6.71	7.87	µg
2-Hexanone	Water	384.10	451.09	µg
2-Methylnaphthalene	Water	0.93	1.09	mg



Source: ISO 14040:2006

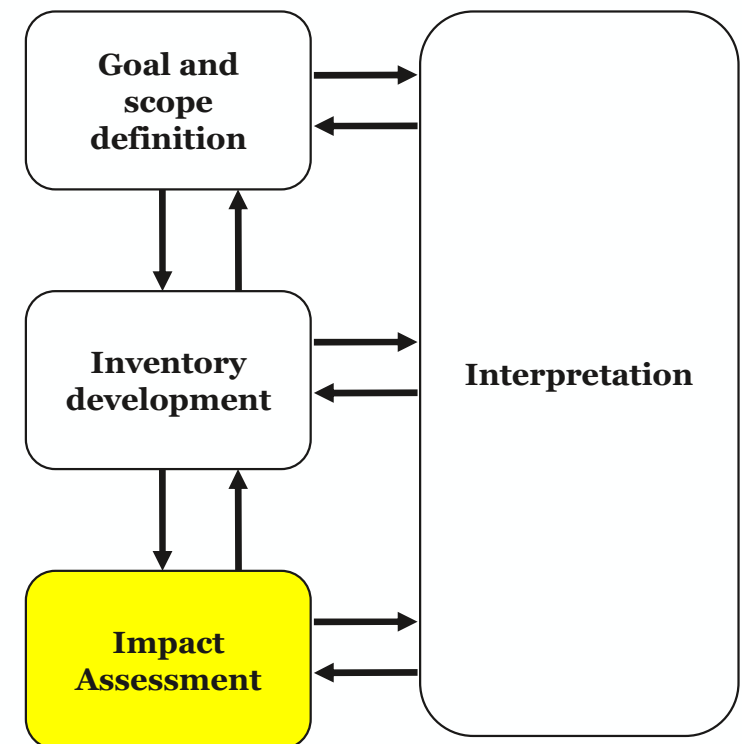
Impact assessment

Must:

- Characterise: calculate impact assessment

Optional:

- Normalise: contribution relative to a known baseline (e.g. national load)
- Weighting of indicators to fewer scores
 - Subjective → need to calculate for different weightings



Source: ISO 14040:2006

Impact assessment - characterisation

Air emission	Flow kg emitted / functional unit	Characterisation factor kg CO ₂ -eq/ kg emitted	Characterised result kg CO ₂ -eq / functional unit
CO ₂	0.1	1	0.1 x 1
CH ₄	0.09	21	0.09 x 21
N ₂ O	0.00012	310	0.00012 x 310
Sum			2.0272

Impact assessment - characterisation

Inventory

Compartment	Elementary flow	Amount (kg/ functional unit)
Raw	Air	0.01563
Raw	Aluminium, 24% in bauxite, 11% in crude ore, in ground	2.11E-02
Raw	Aluminum hydroxide	2.11E-02
Air	Cyclohexane	3.91E-08
Air	Diethylamine	1.47E-20
Air	Dimethyl malonate	7.86E-22
Air	Dinitrogen monoxide	2.35E-06
etc.		

Indicator 1

Compartment	Elementary flow	Characterisation factor (\$/kg)
Raw	Coal, 13.3 MJ per kg, in ground	0.01563
Raw	Coal, 18 MJ per kg, in ground	2.11E-02
Raw	Coal, 18.0 MJ per kg, in ground	2.11E-02
etc.		

Indicator 2

Compartment	Elementary flow	Characterisation factor (kg NMVOC/kg)
Air	1-Butanol	1.05
Air	1-Butene	1.82
Air	1-Butene, 2-methyl-	1.3
etc.		

Interpretation

Iterate back to other steps to:

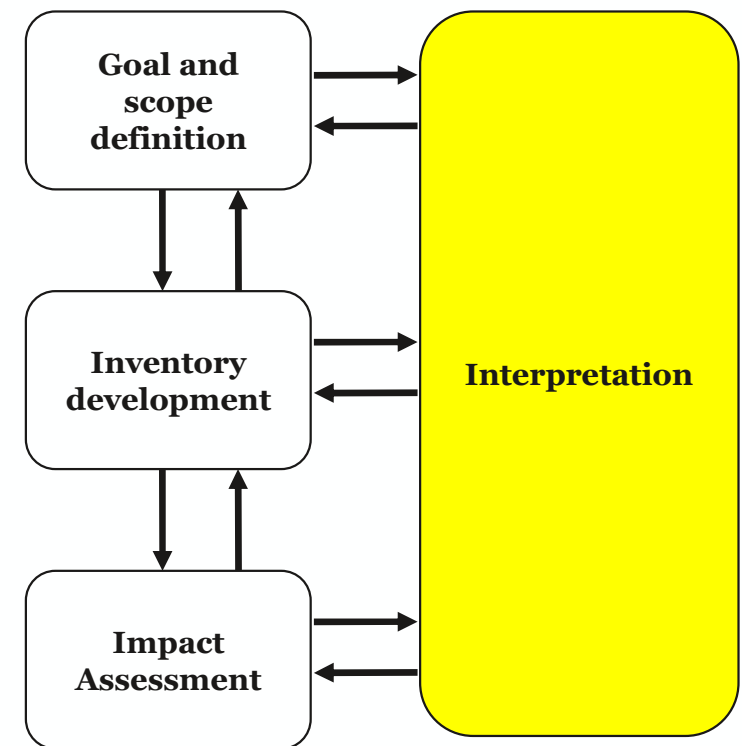
- Identify the reasons for your results → ‘5 whys?’
- Check methods inc. assumptions, data quality

Sensitivity studies

- The effect of key assumptions on your outcomes

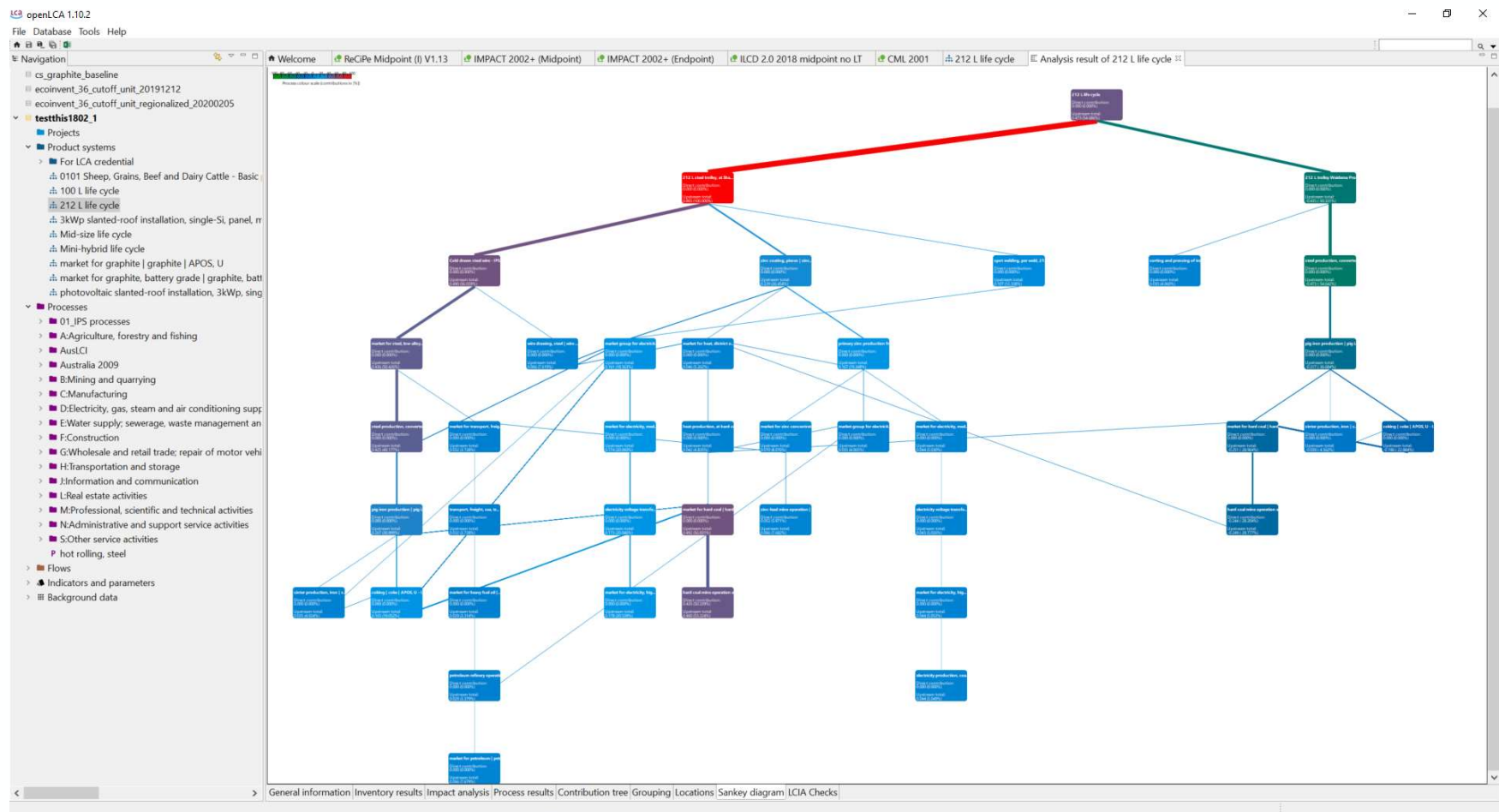
Uncertainty studies

- The effect of data uncertainty on outcomes → confidence in your claims
- Monte Carlo simulations



Source: ISO 14040:2006

Interpretation



Practicalities of LCA

openLCA modeling suite | openLCA.org

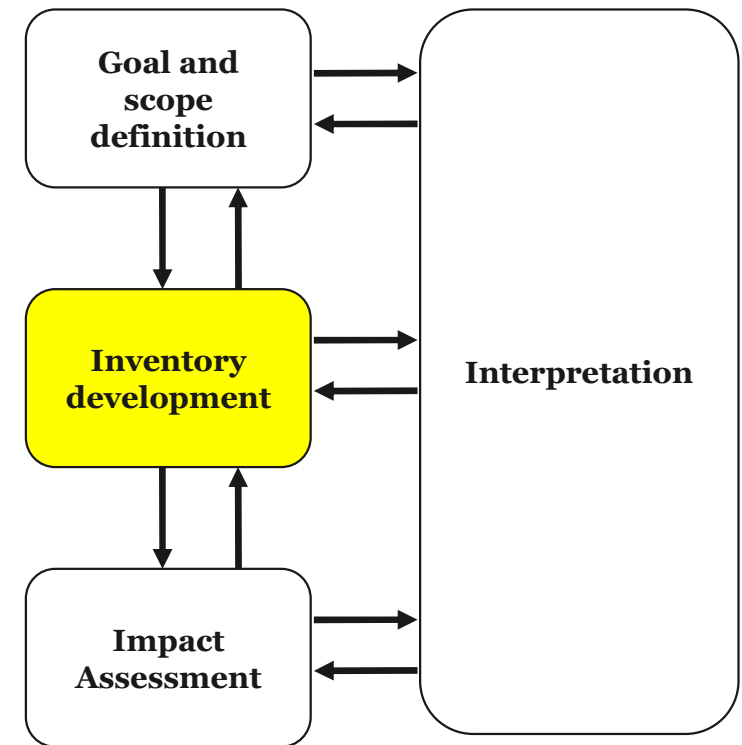
Factor	Streamlined LCA Basic	Streamlined LCA with tool-support	Full LCA
Typical use	Get a feel for the issues	Quick decisions, single product or comparisons	Detailed decisions or comparative claims
Standard compliance	None	Typically aligned to ISO 14040/14044	ISO 14040 / 14044 (or other international standard)
Data	Generic, publicly available	Generic, limited dataset	Supply-chain specific
Software	Spreadsheets	Online or integrated tools	OpenLCA, Sphera (GaBi), SimaPro
Specificity	Low	Medium	High to very high
Uncertainty	Very high	Medium to High	Low
Expertise level needed	Low	Medium to high	High to very high
Time taken	Hours	Days to weeks	Months to years
Cost (for software + expertise)	Just your time	\$100s → \$1000s	> \$10 k



Your assignment based at this level

Full LCA

- ■ 19:Manufacture of coke and refined petroleum products
- ■ 20:Manufacture of chemicals and chemical products
- ■ 21:Manufacture of basic pharmaceutical products and pharmaceutical preparations
- ■ 22:Manufacture of rubber and plastics products
- ■ 23:Manufacture of other non-metallic mineral products
- ■ 24:Manufacture of basic metals
- ■ 25:Manufacture of fabricated metal products, except machinery and equipment
- ■ 26:Manufacture of computer, electronic and optical products
 - ▼ ■ 261:Manufacture of electronic components and boards
 - ▼ ■ 2610:Manufacture of electronic components and boards
 - P cable production, connector for computer, without plugs | cable, connector for computer, without plugs | Cutoff, U - GLO
 - P cable production, data cable in infrastructure | cable, data cable in infrastructure | Cutoff, U - GLO
 - P cable production, network cable, category 5, without plugs | cable, network cable, category 5, without plugs | Cutoff, U - GLO
 - P cable production, printer cable, without plugs | cable, printer cable, without plugs | Cutoff, U - GLO
 - P cable production, ribbon cable, 20-pin, with plugs | cable, ribbon cable, 20-pin, with plugs | Cutoff, U - GLO
 - P capacitor production, auxiliaries and energy use | capacitor, auxiliaries and energy use | Cutoff, U - GLO
 - P capacitor production, electrolyte type, < 2cm height | capacitor, electrolyte type, < 2cm height | Cutoff, U - GLO
 - P capacitor production, electrolyte type, > 2cm height | capacitor, electrolyte type, > 2cm height | Cutoff, U - GLO
 - P capacitor production, film type, for through-hole mounting | capacitor, film type, for through-hole mounting | Cutoff, U - GLO
 - P capacitor production, for surface-mounting | capacitor, for surface-mounting | Cutoff, U - GLO
 - P capacitor production, tantalum-, for through-hole mounting | capacitor, tantalum-, for through-hole mounting | Cutoff, U - GLO
 - P charger production, for electric scooter | charger, for electric scooter | Cutoff, U - GLO
 - P controller production, for electric scooter | controller, for electric scooter | Cutoff, U - GLO
 - P diode production, auxiliaries and energy use | diode, auxiliaries and energy use | Cutoff, U - GLO
 - P diode production, glass-, for surface-mounting | diode, glass-, for surface-mounting | Cutoff, U - GLO
 - P diode production, glass-, for through-hole mounting | diode, glass-, for through-hole mounting | Cutoff, U - GLO
 - P electric connector production, peripheral component interconnect buss | electric connector, peripheral component interconnect buss | Cutoff, U - GLO
 - P electric connector production, peripheral type buss | electric connector, peripheral type buss | Cutoff, U - GLO
 - P electric connector production, wire clamp | electric connector, wire clamp | Cutoff, U - GLO
 - P electric motor production, for electric scooter | electric motor, for electric scooter | Cutoff, U - GLO
 - P electron gun production, for cathode ray tube display | electron gun, for cathode ray tube display | Cutoff, U - GLO
 - P electronic component production, active, unspecified | electronic component, active, unspecified | Cutoff, U - GLO
 - P electronic component production, passive, unspecified | electronic component, passive, unspecified | Cutoff, U - GLO
 - P fan production, for power supply unit, desktop computer | fan, for power supply unit, desktop computer | Cutoff, U - GLO
 - P inductor production, auxiliaries and energy use | inductor, auxiliaries and energy use | Cutoff, U - GLO
 - P inductor production, low value multilayer chip | inductor, low value multilayer chip | Cutoff, U - GLO
 - P inductor production, miniature radio frequency chip | inductor, miniature radio frequency chip | Cutoff, U - GLO
 - P inductor production, ring core choke type | inductor, ring core choke type | Cutoff, U - GLO
 - P insulated gate bipolar transistor production, for electric vehicle applications | insulated gate bipolar transistor, electric vehicle application | Cutoff, U - GLO
 - P integrated circuit production, logic type | integrated circuit, logic type | Cutoff, U - GLO
 - P integrated circuit production, memory type | integrated circuit, memory type | Cutoff, U - GLO
 - P light emitting diode production | light emitting diode | Cutoff, U - GLO
 - P liquid crystal display production, unmounted | liquid crystal display, unmounted | Cutoff, U - GLO
 - P market for cable, connector for computer, without plugs | cable, connector for computer, without plugs | Cutoff, U - GLO
 - P market for cable, data cable in infrastructure | cable, data cable in infrastructure | Cutoff, U - GLO
 - P market for cable, network cable, category 5, without plugs | cable, network cable, category 5, without plugs | Cutoff, U - GLO
 - P market for cable, printer cable, without plugs | cable, printer cable, without plugs | Cutoff, U - GLO
 - P market for cable, ribbon cable, 20-pin, with plugs | cable, ribbon cable, 20-pin, with plugs | Cutoff, U - GLO
 - P market for capacitor, auxiliaries and energy use | capacitor, auxiliaries and energy use | Cutoff, U - GLO



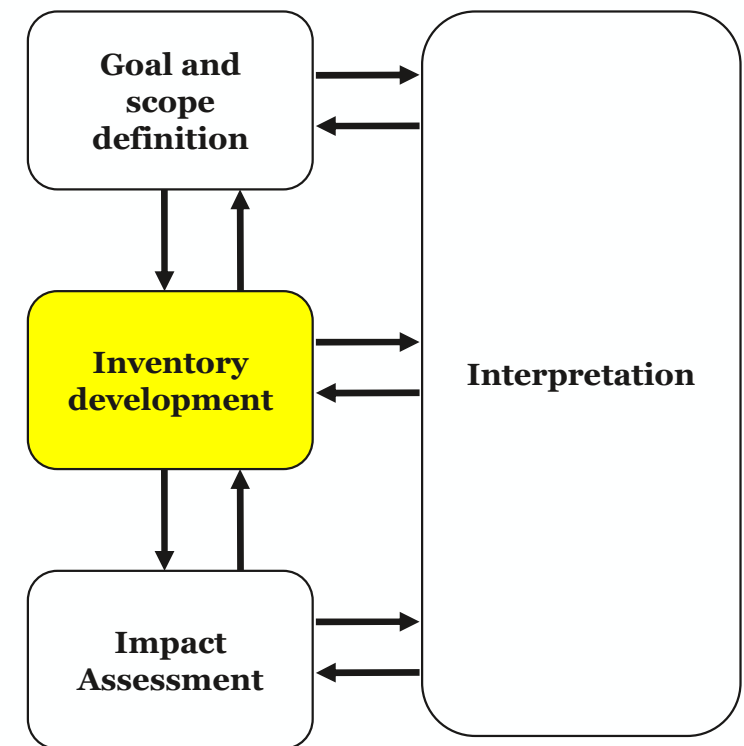
Source: ISO 14040:2006

Full LCA

Inputs/Outputs: electric motor production, for electric scooter | electric motor, for electric scooter | Cutoff, U

Inputs					
Flow	Category	Amount	Unit	Costs/Revenues	Uncertainty
fr aluminium scrap, new	242:Manufacture of basic precious and oth...	-0.08252	kg	-0.06651 EUR	lognormal: gmean=-...
fr aluminium, cast alloy	242:Manufacture of basic precious and oth...	0.41259	kg		lognormal: gmean=0...
fr copper	072:Mining of non-ferrous metal ores/072...	0.17500	kg		lognormal: gmean=0...
fr copper scrap, sorted, pressed	242:Manufacture of basic precious and oth...	-0.03500	kg	-0.09940 EUR	lognormal: gmean=-...
fr electricity, medium voltage	351:Electric power generation, transmission...	2.97500	kWh		lognormal: gmean=2...
fr heat, district or industrial, natural gas	351:Electric power generation, transmission...	6.44000	MJ		lognormal: gmean=6...
fr iron scrap, unsorted	383:Materials recovery/3830:Materials reco...	-0.09759	kg	-0.01454 EUR	lognormal: gmean=-...
fr permanent magnet, for electric motor	242:Manufacture of basic precious and oth...	0.05500	kg		none
fr polyphenylene sulfide	201:Manufacture of basic chemicals, fertiliz...	0.03293	kg		lognormal: gmean=0...
fr steel, low-alloyed, hot rolled	241:Manufacture of basic iron and steel/24...	0.56259	kg		lognormal: gmean=0...
Outputs					
Flow	Category	Amount	Unit	Costs/Revenues	Uncertainty
fr electric motor, for electric scooter	261:Manufacture of electronic compon...	1.00000	kg	143.00000 EUR	none
fr waste plastic, industrial electronics	382:Waste treatment and disposal/3821:Re...	4.46478E-5	kg		lognormal: gmean=-...
fr waste plastic, industrial electronics	382:Waste treatment and disposal/3821:Re...	0.00296	kg		lognormal: gmean=-...

- Each process input/output would be checked
- Modifications needed to reflect actual supply chain



Source: ISO 14040:2006

Full LCA

- Computation of elementary flows associated with the interconnected system

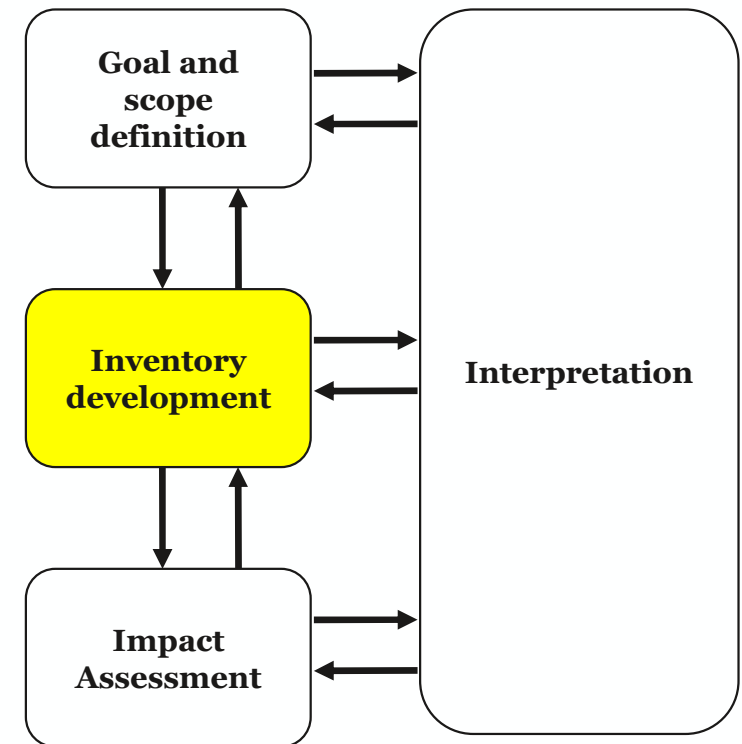
electric motor production, for electric scooter | electric motor, for electric scooter | Cutoff, U

Inputs

Name	Category	Sub-category	Amount	Unit
Aluminium, in ground	Resource	in ground	0.12975	kg
Antimony, in ground	Resource	in ground	5.31349E-7	kg
Antimony, in ground	Resource	in ground	9.23108E-8	kg
Argon-40	Resource	in air	0.00236	kg
Boron, in ground	Resource	in ground	0.00217	kg
Boron, in ground	Resource	in ground	0.00072	kg
Boron, in ground	Resource	in ground	5.35266E-6	kg
Bromine, in water	Resource	in water	3.02052E-7	kg
Calcium, in ground	Resource	in ground	1.56142E-6	kg
Calcium, in ground	Resource	in ground	0.33206	kg
Calcium, in ground	Resource	in ground	0.00165	kg

Outputs

Name	Category	Sub-category	Amount	Unit
1,3-Dioxolane-2-one	Emission to water	unspecified	4.05224E-7	kg
1,4-Butanediol	Emission to air	high population density	3.86434E-7	kg
1,4-Butanediol	Emission to water	surface water	8.88799E-7	kg
1-Pentanol	Emission to water	surface water	2.55779E-11	kg
1-Pentanol	Emission to air	high population density	1.06573E-11	kg
1-Pentene	Emission to water	surface water	1.93288E-11	kg
1-Pentene	Emission to air	high population density	6.45778E-11	kg
2,2,4-Trimethyl pentane	Emission to air	unspecified	9.86344E-13	kg
2,4-D	Emission to air	low population density	1.34599E-9	kg
2,4-D	Emission to soil	agricultural	8.32174E-8	kg
2,4-D amine	Emission to air	low population density	9.18966E-10	kg

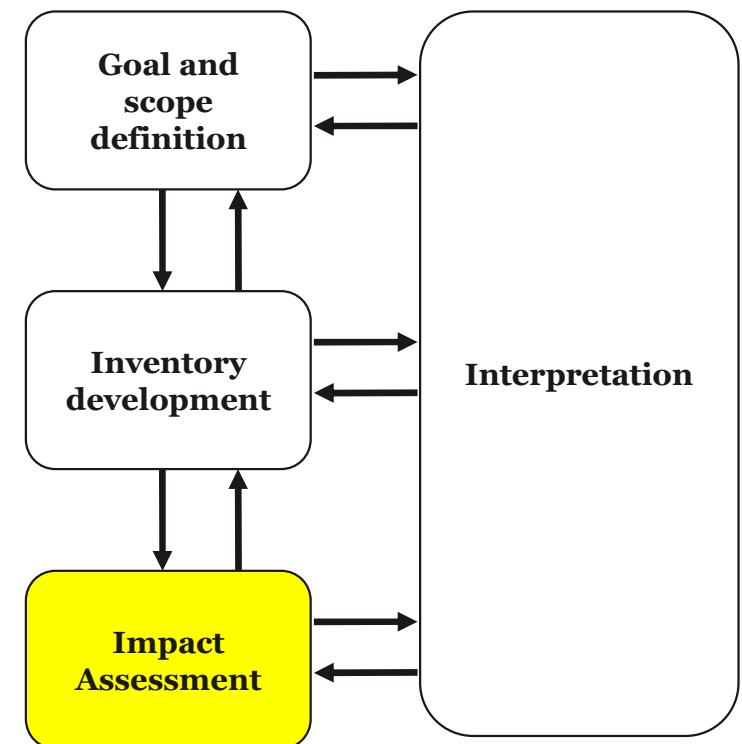


Source: ISO 14040:2006

Full LCA

- Translation of LCI flows into environmental impacts

Name	Impact result	Unit
> A Greenhouse Gas - Total	8.03606	kg CO2 eq
> B Water	0.51054	m3
> C Marine acidification, short term	0.11687	PDF.m2.yr
> D Marine acidification, long term	1.07662	PDF.m2.yr
> E Freshwater acidification	0.22813	PDF.m2.yr
> F Freshwater eutrophication	0.00242	PDF.m2.yr
> G Marine eutrophication	0.01458	PDF.m2.yr
> H Freshwater ecotoxicity, short term	0.33985	PDF.m2.yr
> I Freshwater ecotoxicity, long term	1873.31450	PDF.m2.yr
> J Land occupation	0.09592	m2org.arable



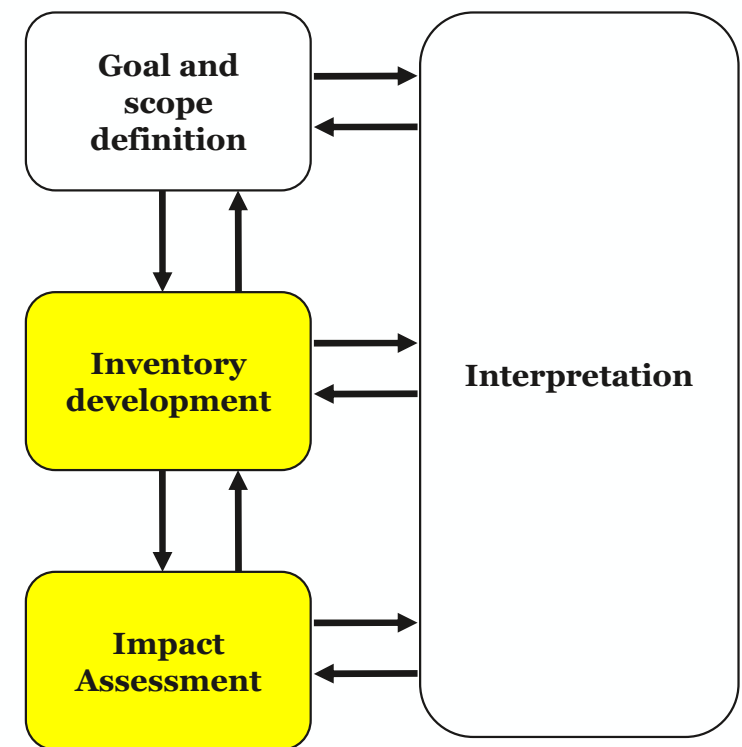
Source: ISO 14040:2006

Streamlined LCA – with tool support

- LCI and LCIA steps are combined
- ‘Background’ Emission Factors (EFs) used to account for impacts for any process flow inputs/outputs
- May not need to model direct emissions
- Limited customisability
- Full life cycle or just production?

Example software:

- oneclicklca.com
- Sustainable Minds
- Ecodex



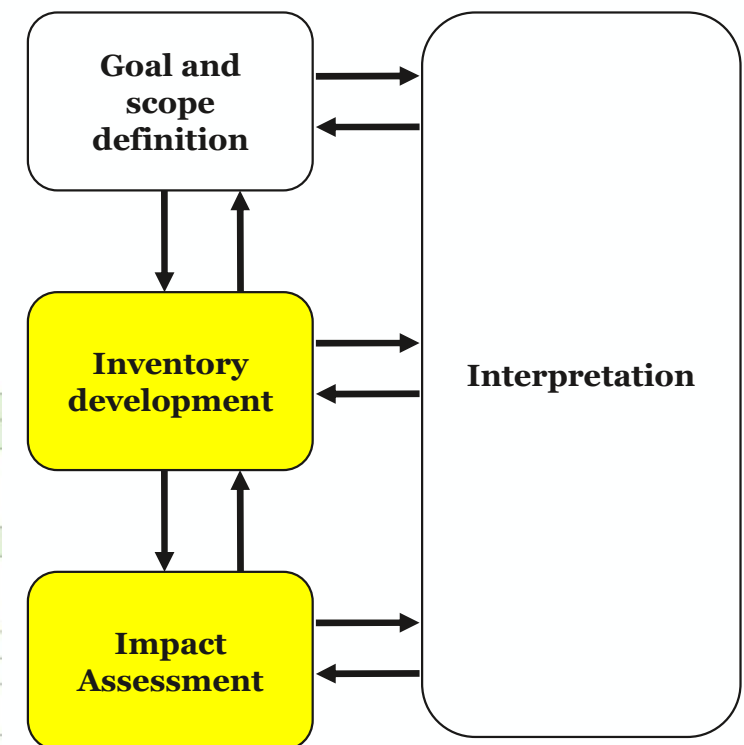
Source: ISO 14040:2006

Streamlined LCA - basic

Inventory of Carbon and Energy

- <https://circularecology.com/embodied-carbon-footprint-database.html>
- Emission factors for material production
- Shaping processes may or may not be included

Materials	Embodied Carbon Kg CO2/sqm
PV Modules	
Monocrystalline	242 (132 to 440) CO2 Only
Polycrystalline	208 (99 to 289) CO2 Only
Thin Film	67 (40 to 92) CO2 Only
Roads	
Asphalt road - Hot construction method - 40 yrs	
Construction	32.8 KgCO2/Sqm
Maintenance - 40 yrs	12.3 KgCO2/Sqm
Operation - 40 yrs	54.0 KgCO2/Sqm
Asphalt road - Cold construction method - 40 yrs	
Construction	28.2 KgCO2/Sqm
Maintenance - 40 yrs	14.8 KgCO2/Sqm
Operation - 40 yrs	54.0 KgCO2/Sqm



Source: ISO 14040:2006

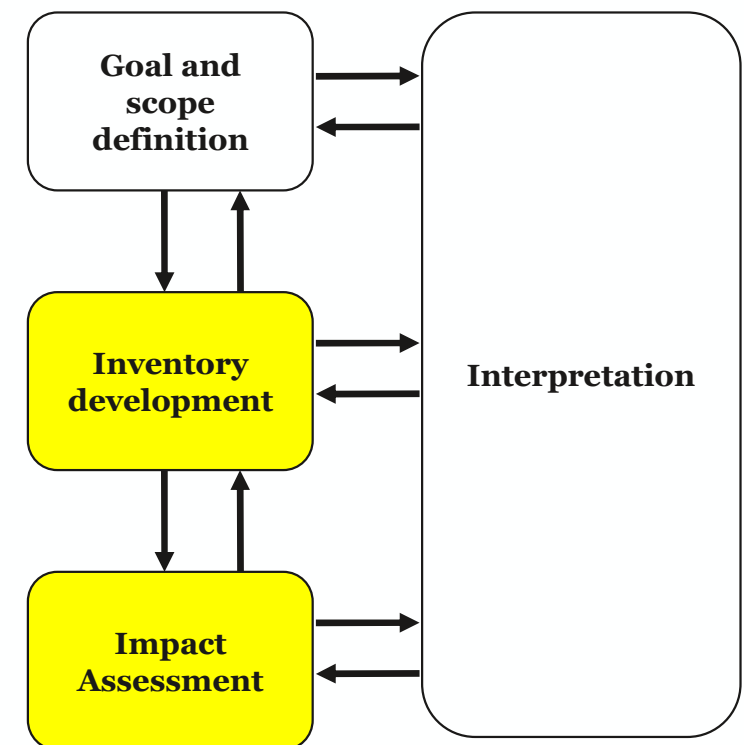
Streamlined LCA - basic

- Emission factors for electricity / fuel use / transport
- Ministry for the Environment | Manatū Mō Te Taiao
- <https://environment.govt.nz/assets/Publications/Files/Measuring-Emissions-Detailed-Guide-2020.pdf>

Emission source	Unit	kg CO ₂ -e/unit
Residential use		
Coal – default	kg	1.88
Coal – bituminous	kg	2.86
Coal – sub-bituminous	kg	2.15
Coal – lignite	kg	1.54

Emission source	Unit	kg CO ₂ -e/unit
Rail freight	tkm	0.028

Emission source	Unit	kg CO ₂ -e/unit
2018	kWh	0.101
2017	kWh	0.113



Source: ISO 14040:2006

Streamlined LCA - basic

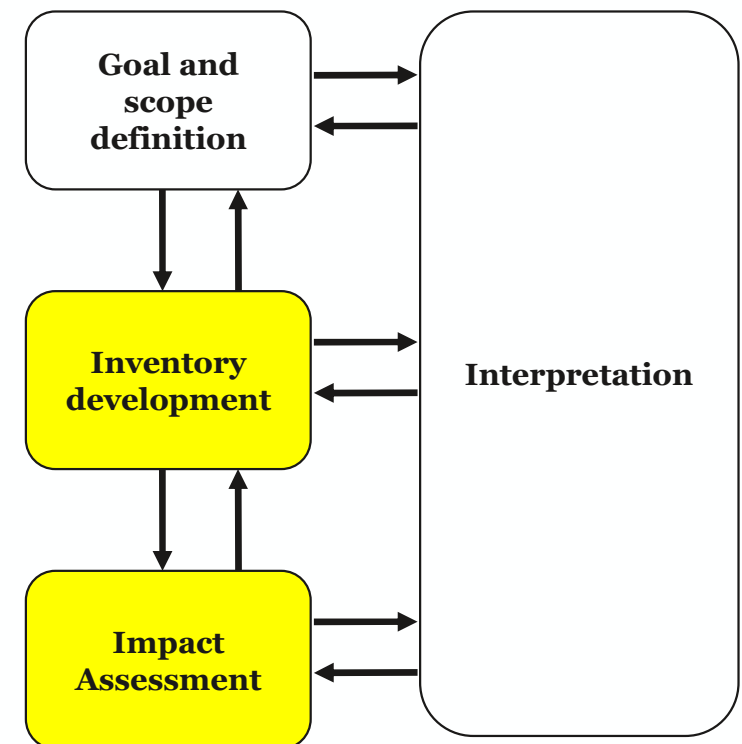
$$E = Q \times F$$

E = Emissions from the emissions source in kg CO₂-eq

Q = Activity data, e.g. quantity of material used

F = Emission factor for emissions source

1. Set functional unit, system boundary
2. Collect data
3. Use $E = Q \times F$ in a spreadsheet, add everything up
4. Produce a report



Source: ISO 14040:2006

Concluding remarks

- Spend time on the project scope, inc. what you are trying to answer
 - How will you use the LCA with your environmental strategy?
- Understand what has been done before and what data is available
- Never assume your product or service is green
- Stakeholders engagement is critical